

12 LINEAR REGRESSION AND CORRELATION

Figure 12.1 Linear regression and correlation can help you determine if an auto mechanic's salary is related to his work experience. (credit: modification of work "USPS commissions local repair-shop for some needed work on its older trucks" by Joshua Rothhaas/ Flickr, CC BY 2.0)

Chapter Objectives

By the end of this chapter, the student should be able to:

- Discuss basic ideas of linear regression and correlation.
- Create and interpret a line of best fit.
- Calculate and interpret the correlation coefficient.
- Calculate and interpret outliers.

Introduction

Professionals often want to know how two or more numeric variables are related. For example, is there a relationship between the grade on the second math exam a student takes and the grade on the final exam? If there is a relationship, what is the relationship and how strong is it?

In another example, your income may be determined by your education, your profession, your years of experience, and your ability. The amount you pay a repair person for labor is often determined by an initial amount plus an hourly fee.

The type of data described in the examples is **bivariate** data — "bi" for two variables. In reality, statisticians use **multivariate** data, meaning many variables.

In this chapter, you will be studying the simplest form of regression, "linear regression" with one independent variable (x). This involves data that fits a line in two dimensions. You will also study correlation which measures how strong the relationship is.

12.1 Linear Equations

Linear regression for two variables is based on a linear equation with one independent variable. The equation has the form:

$$y = a + \mathrm{bx}$$

where a and b are constant numbers.

The variable ***x* is the independent variable, and *y* is the dependent variable.** Typically, you choose a value to substitute for the independent variable and then solve for the dependent variable.

EXAMPLE 12.1

The following examples are linear equations.

$$y = 3 + 2x$$
$$y = -0.01 + 1.2x$$

TRY IT 12.1

Is the following an example of a linear equation?

$y = -0.125 - 3.5x$

The graph of a linear equation of the form $y = a + bx$ is a **straight line**. Any line that is not vertical can be described by this equation.

EXAMPLE 12.2

Graph the equation $y = -1 + 2x$.

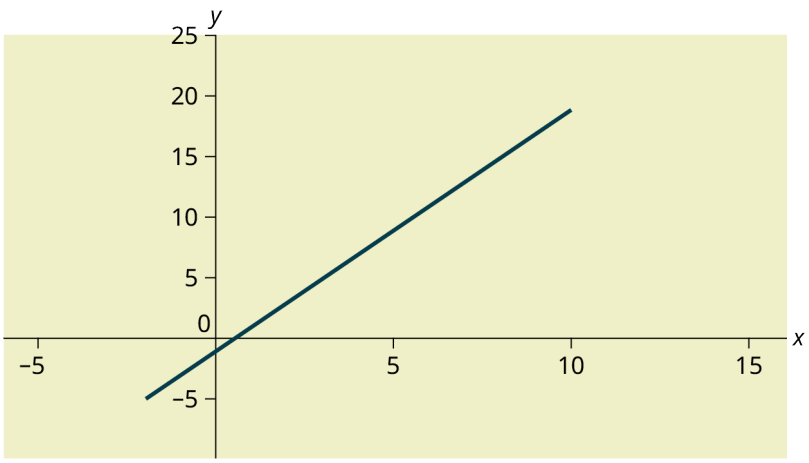


Figure 12.2

TRY IT 12.2

Is the following an example of a linear equation? Why or why not?



Figure 12.3

EXAMPLE 12.3

A local small business completes federal tax returns for customers. The rate for services is $32 per hour plus a $31.50 one-time charge. The total cost to a customer depends on the number of hours it takes to complete the job.

Problem

Find the equation that expresses the **total cost** in terms of the **number of hours** required to complete the job.

Solution

Let x = the number of hours it takes to get the job done.
Let y = the total cost to the customer.

The $31.50 is a fixed cost. If it takes x hours to complete the tax return, then (32)(x) is the cost of the tax return processing only. The total cost is: $y = 31.50 + 32x$

TRY IT 12.3

Emma's Extreme Sports hires hang-gliding instructors and pays them a fee of $50 per class as well as $20 per student in the class. The total cost Emma pays depends on the number of students in a class. Find the equation that expresses the total cost in terms of the number of students in a class.

Slope and *Y*-Intercept of a Linear Equation

For the linear equation $y = a + bx$, b = slope and a = y-intercept. From algebra recall that the slope is a number that describes the steepness of a line, and the y-intercept is the y coordinate of the point (0, a) where the line crosses the y-axis.



(a) (b) (c)

Figure 12.4 Three possible graphs of $y = a + bx$. (a) If $b > 0$, the line slopes upward to the right. (b) If $b = 0$, the line is horizontal. (c) If $b < 0$, the line slopes downward to the right.

EXAMPLE 12.4

Svetlana tutors to make extra money for college. For each tutoring session, she charges a one-time fee of $25 plus $15 per hour of tutoring. A linear equation that expresses the total amount of money Svetlana earns for each session she tutors is $y = 25 + 15x$.

Problem

What are the independent and dependent variables? What is the y-intercept and what is the slope? Interpret them using complete sentences.

Solution

The independent variable (x) is the number of hours Svetlana tutors each session. The dependent variable (y) is the amount, in dollars, Svetlana earns for each session.

The y-intercept is 25 ($a = 25$). At the start of the tutoring session, Svetlana charges a one-time fee of $25 (this is when $x = 0$). The slope is 15 ($b = 15$). For each session, Svetlana earns $15 for each hour she tutors.

TRY IT 12.4

Ethan repairs household appliances like dishwashers and refrigerators. For each visit, he charges $25 plus $20 per hour of work. A linear equation that expresses the total amount of money Ethan earns per visit is $y = 25 + 20x$.

What are the independent and dependent variables? What is the y-intercept and what is the slope? Interpret them using complete sentences.

12.2 Scatter Plots

Before we take up the discussion of linear regression and correlation, we need to examine a way to display the relation between two variables x and y. The most common and easiest way is a **scatter plot**. The following example illustrates a scatter plot.

EXAMPLE 12.5

An educational researcher collects data on the vocabulary size of children as a function of age. The data is shown in Table 12.1. Is there a relationship between age and vocabulary size for young children? Construct a scatter plot. Let x = Child's Age, and let y = Vocabulary Size.

Age (years)	Vocabulary Size (number of words)
3	655
4	1098
6	2463
7	3195

Table 12.1

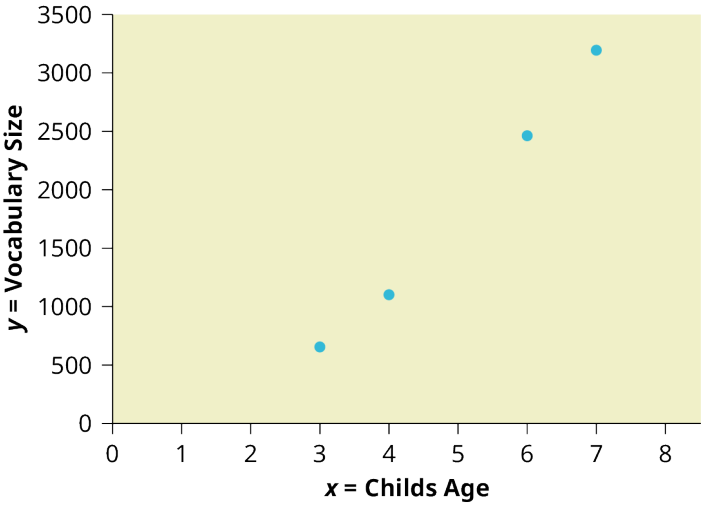


Figure 12.5

USING THE TI-83, 83+, 84, 84+ CALCULATOR

To create a scatter plot:

1. Enter your X data into list L1 and your Y data into list L2.
2. Press 2nd STATPLOT ENTER to use Plot 1. On the input screen for PLOT 1, highlight On and press ENTER. (Make sure the other plots are OFF.)
3. For TYPE: highlight the very first icon, which is the scatter plot, and press ENTER.
4. For Xlist:, enter L1 ENTER and for Ylist: L2 ENTER.
5. For Mark: it does not matter which symbol you highlight, but the square is the easiest to see. Press ENTER.
6. Make sure there are no other equations that could be plotted. Press Y = and clear any equations out.
7. Press the ZOOM key and then the number 9 (for menu item "ZoomStat") ; the calculator will fit the window to the data. You can press WINDOW to see the scaling of the axes.

TRY IT 12.5

Amelia plays basketball for her high school. She wants to improve to play at the college level. She notices that the number of points she scores in a game goes up in response to the number of hours she practices her jump shot each week. She records the following data:

X (hours practicing jump shot)	*Y* (points scored in a game)
5	15
7	22
9	28
10	31

Table 12.2

X (hours practicing jump shot)	*Y* (points scored in a game)
11	33
12	36

Table 12.2

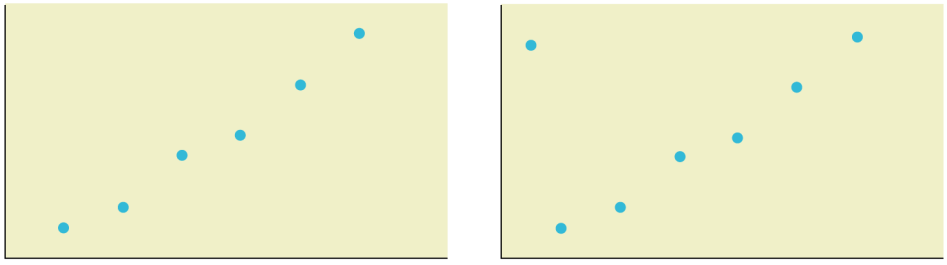
Construct a scatter plot and state if what Amelia thinks appears to be true.

A scatter plot shows the **direction** of a relationship between the variables. A clear direction happens when there is either:

- High values of one variable occurring with high values of the other variable or low values of one variable occurring with low values of the other variable.
- High values of one variable occurring with low values of the other variable.

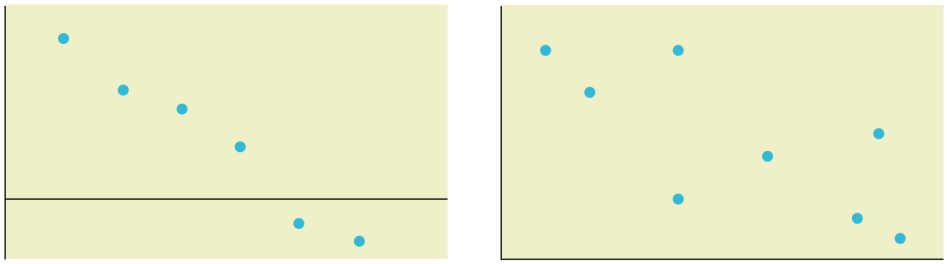
You can determine the **strength** of the relationship by looking at the scatter plot and seeing how close the points are to a line, a power function, an exponential function, or to some other type of function. For a linear relationship there is an exception. Consider a scatter plot where all the points fall on a horizontal line providing a "perfect fit." The horizontal line would in fact show no relationship.

When you look at a scatterplot, you want to notice the **overall pattern** and any **deviations** from the pattern. The following scatterplot examples illustrate these concepts.



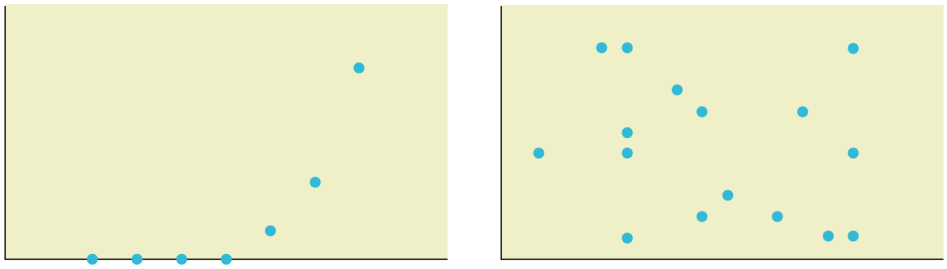
(a) Positive linear pattern (strong) (b) Linear pattern w/ one deviation

Figure 12.6



(a) Negative linear pattern (strong) (b) Negative linear pattern (weak)

Figure 12.7



(a) Exponential growth pattern (b) No pattern

Figure 12.8

In this chapter, we are interested in scatter plots that show a linear pattern. Linear patterns are quite common. The linear relationship is strong if the points are close to a straight line, except in the case of a horizontal line where there is no relationship. If we think that the points show a linear relationship, we would like to draw a line on the scatter plot. This line can be calculated through a process called **linear regression**. However, we only calculate a regression line if one of the variables helps to explain or predict the other variable. If *x* is the independent variable and *y* the dependent variable, then we can use a regression line to predict *y* for a given value of *x*

12.3 The Regression Equation

Data rarely fit a straight line exactly. Usually, you must be satisfied with rough predictions. Typically, you have a set of data whose scatter plot appears to **"fit"** a straight line. This is called a **Line of Best Fit or Least-Squares Line**.

COLLABORATIVE EXERCISE

If you know a person's pinky (smallest) finger length, do you think you could predict that person's height? Collect data from your class (pinky finger length, in inches). The independent variable, *x*, is pinky finger length and the dependent variable, *y*, is height. For each set of data, plot the points on graph paper. Make your graph big enough and **use a ruler**. Then "by eye" draw a line that appears to "fit" the data. For your line, pick two convenient points and use them to find the slope of the line. Find the *y*-intercept of the line by extending your line so it crosses the *y*-axis. Using the slopes and the *y*-intercepts, write your equation of "best fit." Do you think everyone will have the same equation? Why or why not? According to your equation, what is the predicted height for a pinky length of 2.5 inches?

EXAMPLE 12.6

A random sample of 11 statistics students produced the following data, where *x* is the third exam score out of 80, and *y* is the final exam score out of 200. Can you predict the final exam score of a random student if you know the third exam score?

x (third exam score)	y (final exam score)
65	175
67	133
71	185
71	163
66	126

Table 12.3 Table showing the scores on the final exam based on scores from the third exam.

x (third exam score)	y (final exam score)
75	198
67	153
70	163
71	159
69	151
69	159

Table 12.3 Table showing the scores on the final exam based on scores from the third exam.

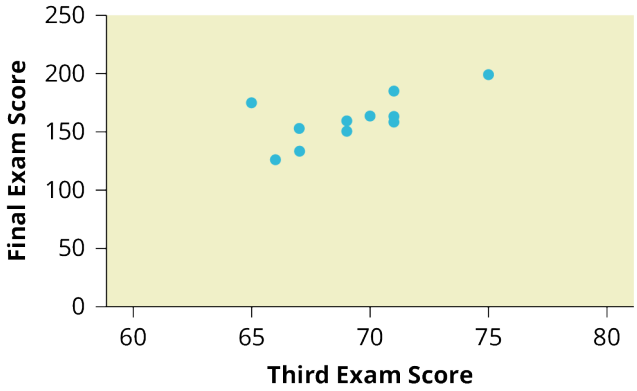


Figure 12.9 Scatter plot showing the scores on the final exam based on scores from the third exam.

TRY IT 12.6

SCUBA divers have maximum dive times they cannot exceed when going to different depths. The data in Table 12.4 show different depths with the maximum dive times in minutes. Use your calculator to find the least squares regression line and predict the maximum dive time for 110 feet.

X (depth in feet)	*Y* (maximum dive time)
50	80
60	55
70	45
80	35

Table 12.4

X (depth in feet)	Y (maximum dive time)
90	25
100	22

Table 12.4

The third exam score, *x*, is the independent variable and the final exam score, *y*, is the dependent variable. We will plot a regression line that best "fits" the data. If each of you were to fit a line "by eye," you would draw different lines. We can use what is called a **least-squares regression line** to obtain the best fit line.

Consider the following diagram. Each point of data is of the the form (*x*, *y*) and each point of the line of best fit using least-squares linear regression has the form (*x*, $\hat{y}$).

The $\hat{y}$ is read **"*y* hat"** and is the **estimated value of *y***. It is the value of *y* obtained using the regression line. It is not generally equal to *y* from data.

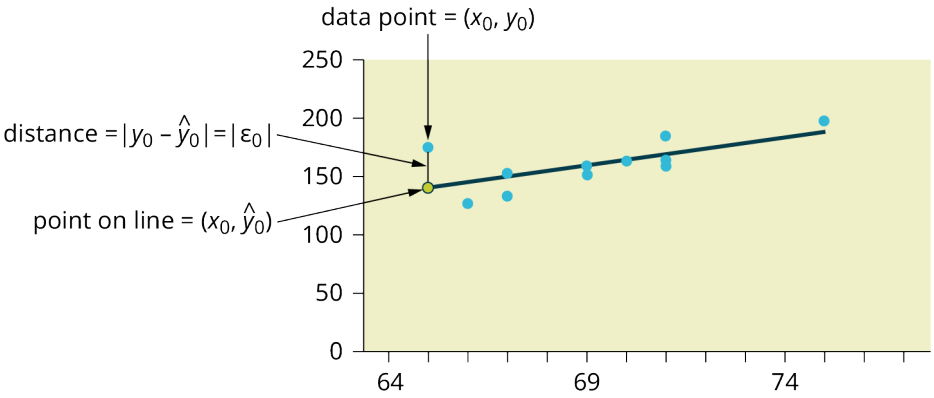


Figure 12.10

The term $y_0 - \hat{y}_0 = \varepsilon_0$ is called the **"error" or residual**. It is not an error in the sense of a mistake. The **absolute value of a residual** measures the vertical distance between the actual value of *y* and the estimated value of *y*. In other words, it measures the vertical distance between the actual data point and the predicted point on the line.

If the observed data point lies above the line, the residual is positive, and the line underestimates the actual data value for *y*. If the observed data point lies below the line, the residual is negative, and the line overestimates that actual data value for *y*.

In the diagram in Figure 12.10, $y_0 - \hat{y}_0 = \varepsilon_0$ is the residual for the point shown. Here the point lies above the line and the residual is positive.

ε = the Greek letter **epsilon**

For each data point, you can calculate the residuals or errors, $y_i - \hat{y}_i = \varepsilon_i$ for $i = 1, 2, 3, \ldots, 11$.

Each $|\varepsilon|$ is a vertical distance.

For the example about the third exam scores and the final exam scores for the 11 statistics students, there are 11 data points. Therefore, there are 11 ε values. If you square each ε and add, you get

$$(\varepsilon_1)^2 + (\varepsilon_2)^2 + \ldots + (\varepsilon_{11})^2 = \sum_{i=1}^{11} \varepsilon^2$$

This is called the **Sum of Squared Errors (SSE)**.

Using calculus, you can determine the values of *a* and *b* that make the **SSE** a minimum. When you make the **SSE** a minimum, you have determined the points that are on the line of best fit. It turns out that the line of best fit has the equation:

$$\hat{y} = a + bx$$

where $a = \overline{y} - b\overline{x}$ and $b = \frac{\Sigma(x-\overline{x})(y-\overline{y})}{\Sigma(x-\overline{x})^2}$.

The sample means of the x values and the y values are $\overline{x}$ and $\overline{y}$, respectively. The best fit line always passes through the point $(\overline{x}, \overline{y})$.

The slope b can be written as $b = r\left(\frac{s_y}{s_x}\right)$ where s_y = the standard deviation of the y values and s_x = the standard deviation of the x values. r is the correlation coefficient, which is discussed in the next section.

Residuals Plots

A residuals plot can be used to help determine if a set of (x, y) data is linearly correlated. For each data point used to create the correlation line, a residual $y - \hat{y}$ can be calculated, where y is the observed value of the response variable and $\hat{y}$ is the value predicted by the correlation line. The difference between these values is called the residual. A residuals plot shows the explanatory variable x on the horizontal axis and the residual for that value on the vertical axis. The residuals plot is often shown together with a scatter plot of the data. While a scatter plot of the data should resemble a straight line, a residuals plot should appear random, with no pattern and no outliers. It should also show constant error variance, meaning the residuals should not consistently increase (or decrease) as the explanatory variable x increases.

A residuals plot can be created using StatCrunch or a TI calculator. The plot should appear random. A box plot of the residuals is also helpful to verify that there are no outliers in the data. By observing the scatter plot of the data, the residuals plot, and the box plot of residuals, together with the linear correlation coefficient, we can usually determine if it is reasonable to conclude that the data are linearly correlated.

EXAMPLE:

A shop owner uses a straight-line regression to estimate the number of ice cream cones that would be sold in a day based on the temperature at noon. The owner has data for a 2-year period and chose nine days at random. A scatter plot of the data is shown, together with a residuals plot.

Temperature ° F	Ice cream cones sold
70	105
85	240
65	49
72	147
80	231
61	38
75	193
78	196
68	89

Table 12.5 Table showing the number of ice cream cones sold on nine random days and the temperature at noon on those days.

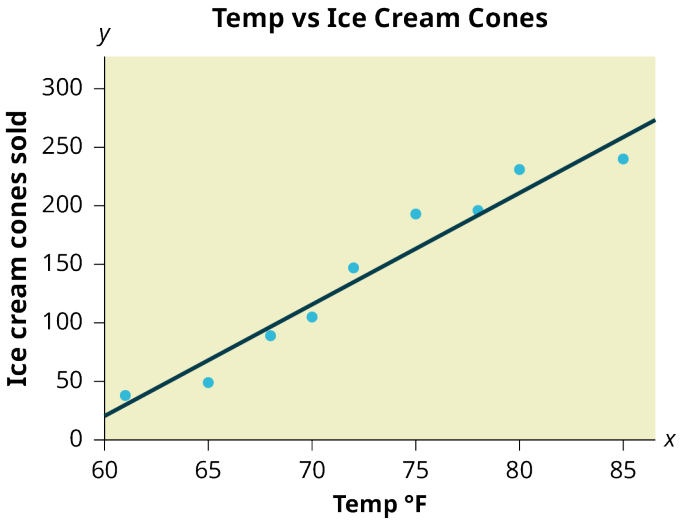


Figure 12.11 Scatter plot of the data looks like a straight line.

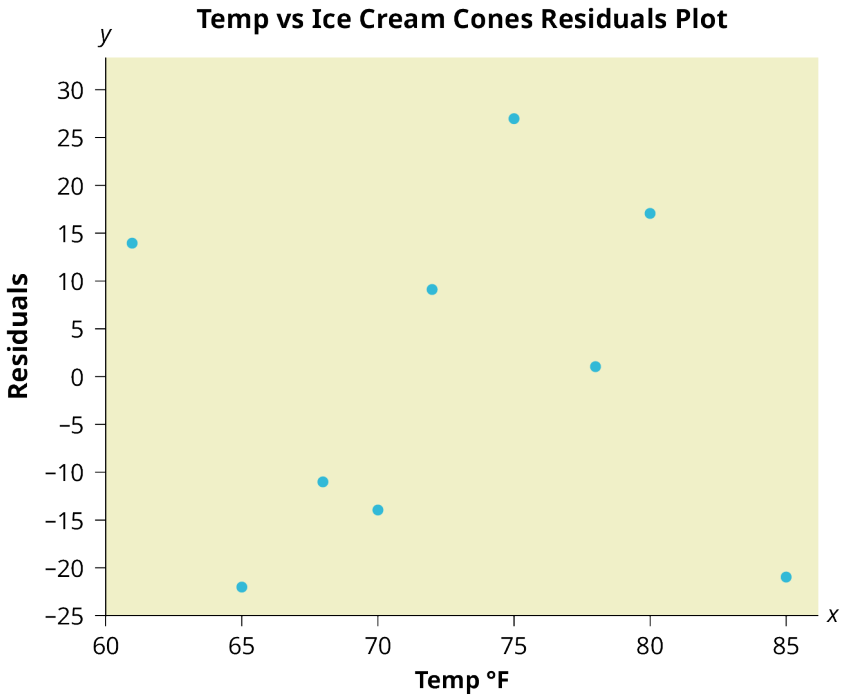


Figure 12.12 Residuals plot appears random.

Least Squares Criteria for Best Fit

The process of fitting the best-fit line is called **linear regression**. The idea behind finding the best-fit line is based on the assumption that the data are scattered about a straight line. The criteria for the best fit line is that the sum of the squared errors (SSE) is minimized, that is, made as small as possible. Any other line you might choose would have a higher SSE than the best fit line. This best fit line is called the **least-squares regression line** .

NOTE

Computer spreadsheets, statistical software, and many calculators can quickly calculate the best-fit line and create the graphs. The calculations tend to be tedious if done by hand. Instructions to use the TI-83, TI-83+, and TI-84+ calculators to find the best-fit line and create a scatterplot are shown at the end of this section.

THIRD EXAM vs FINAL EXAM EXAMPLE:
The graph of the line of best fit for the third-exam/final-exam example is as follows:

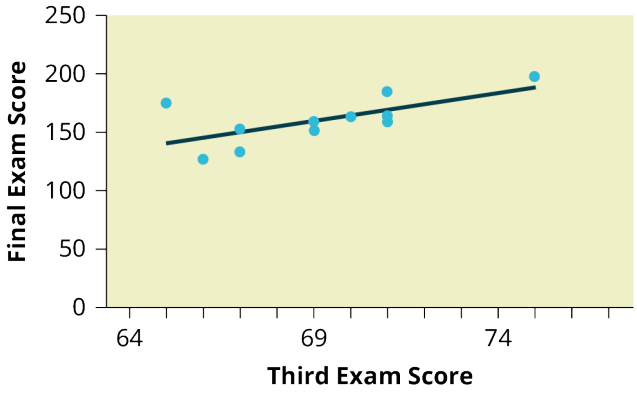


Figure 12.13

The least squares regression line (best-fit line) for the third-exam/final-exam example has the equation:

$$\hat{y} = -173.51 + 4.83x$$

REMINDER

Remember, it is always important to plot a scatter diagram first. If the scatter plot indicates that there is a linear relationship between the variables, then it is reasonable to use a best fit line to make predictions for y given x within the domain of x-values in the sample data, **but not necessarily for x-values outside that domain.** You could use the line to predict the final exam score for a student who earned a grade of 73 on the third exam. You should NOT use the line to predict the final exam score for a student who earned a grade of 50 on the third exam, because 50 is not within the domain of the x-values in the sample data, which are between 65 and 75.

UNDERSTANDING SLOPE

The slope of the line, b, describes how changes in the variables are related. It is important to interpret the slope of the line in the context of the situation represented by the data. You should be able to write a sentence interpreting the slope in plain English.

INTERPRETATION OF THE SLOPE: The slope of the best-fit line tells us how the dependent variable (y) changes for every one unit increase in the independent (x) variable, on average.

THIRD EXAM vs FINAL EXAM EXAMPLE
Slope: The slope of the line is $b = 4.83$.
Interpretation: For a one-point increase in the score on the third exam, the final exam score increases by 4.83 points, on average.

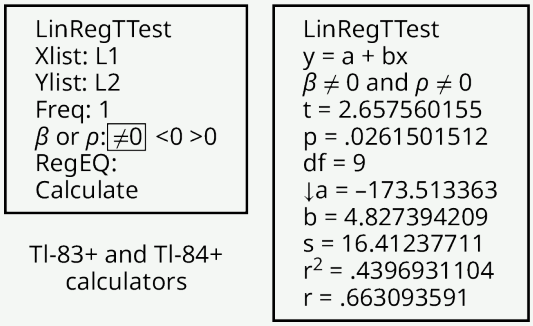
USING THE TI-83, 83+, 84, 84+ CALCULATOR

Using the Linear Regression T Test: LinRegTTest

1. In the STAT list editor, enter the X data in list L1 and the Y data in list L2, paired so that the corresponding (x,y) values are next to each other in the lists. (If a particular pair of values is repeated, enter it as many times as it appears in the data.)
2. On the STAT TESTS menu, scroll down with the cursor to select the LinRegTTest. (Be careful to select LinRegTTest, as some calculators may also have a different item called LinRegTInt.)
3. On the LinRegTTest input screen enter: Xlist: L1 ; Ylist: L2 ; Freq: 1
4. On the next line, at the prompt β or ρ, highlight "$\neq$ 0" and press ENTER
5. Leave the line for "RegEq:" blank

6. Highlight Calculate and press ENTER.

LinRegTTest Input Screen and Output Screen



Tl-83+ and Tl-84+ calculators

Figure 12.14

The output screen contains a lot of information. For now we will focus on a few items from the output, and will return later to the other items.
The second line says $y = a + bx$. Scroll down to find the values $a = -173.513$, and $b = 4.8273$; the equation of the best fit line is $\hat{y} = -173.51 + 4.83x$
The two items at the bottom are $r_2 = 0.43969$ and $r = 0.663$. For now, just note where to find these values; we will discuss them in the next two sections.

Graphing the Scatterplot and Regression Line

1. We are assuming your X data is already entered in list L1 and your Y data is in list L2
2. Press 2nd STATPLOT ENTER to use Plot 1
3. On the input screen for PLOT 1, highlight **On**, and press ENTER
4. For TYPE: highlight the very first icon which is the scatterplot and press ENTER
5. Indicate Xlist: L1 and Ylist: L2
6. For Mark: it does not matter which symbol you highlight.
7. Press the ZOOM key and then the number 9 (for menu item "ZoomStat") ; the calculator will fit the window to the data
8. To graph the best-fit line, press the "Y=" key and type the equation –173.5 + 4.83X into equation Y1. (The X key is immediately left of the STAT key). Press ZOOM 9 again to graph it.
9. Optional: If you want to change the viewing window, press the WINDOW key. Enter your desired window using Xmin, Xmax, Ymin, Ymax

NOTE

Another way to graph the line after you create a scatter plot is to use LinRegTTest.

1. Make sure you have done the scatter plot. Check it on your screen.
2. Go to LinRegTTest and enter the lists.
3. At RegEq: press VARS and arrow over to Y-VARS. Press 1 for 1:Function. Press 1 for 1:Y1. Then arrow down to Calculate and do the calculation for the line of best fit.
4. Press Y = (you will see the regression equation).
5. Press GRAPH. The line will be drawn."

The Correlation Coefficient *r*

Besides looking at the scatter plot and seeing that a line seems reasonable, how can you tell if the line is a good predictor? Use the correlation coefficient as another indicator (besides the scatterplot) of the strength of the relationship between *x* and *y*.

The **correlation coefficient, *r*,** developed by Karl Pearson in the early 1900s, is numerical and provides a measure of strength and direction of the linear association between the independent variable *x* and the dependent variable *y*.

The correlation coefficient is calculated as

$$r = \frac{n\Sigma(xy) - (\Sigma x)(\Sigma y)}{\sqrt{\left[n\Sigma x^2 - (\Sigma x)^2\right]\left[n\Sigma y^2 - (\Sigma y)^2\right]}}$$

where n = the number of data points.

If you suspect a linear relationship between x and y, then r can measure how strong the linear relationship is.

What the VALUE of *r* tells us:

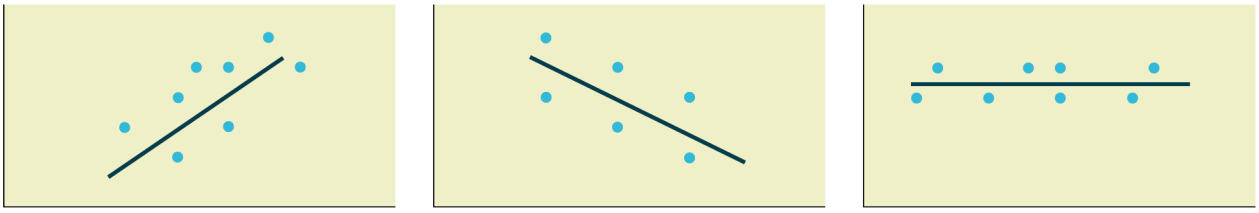
- The value of r is always between –1 and +1: $-1 \leq r \leq 1$.
- The size of the correlation r indicates the strength of the linear relationship between x and y. Values of r close to –1 or to +1 indicate a stronger linear relationship between x and y.
- If $r = 0$ there is likely no linear correlation. It is important to view the scatterplot, however, because data that exhibit a curved or horizontal pattern may have a correlation of 0.
- If $r = 1$, there is perfect positive correlation. If $r = -1$, there is perfect negative correlation. In both these cases, all of the original data points lie on a straight line. Of course,in the real world, this will not generally happen.

What the SIGN of *r* tells us

- A positive value of r means that when x increases, y tends to increase and when x decreases, y tends to decrease **(positive correlation)**.
- A negative value of r means that when x increases, y tends to decrease and when x decreases, y tends to increase **(negative correlation)**.
- The sign of r is the same as the sign of the slope, b, of the best-fit line.

NOTE

Strong correlation does not suggest that x causes y or y causes x. We say **"correlation does not imply causation."**



(a) Positive correlation (b) Negative correlation (c) Zero correlation

Figure 12.15 (a) A scatter plot showing data with a positive correlation. $0 < r < 1$ (b) A scatter plot showing data with a negative correlation. $-1 < r < 0$ (c) A scatter plot showing data with zero correlation. $r = 0$

The formula for r looks formidable. However, computer spreadsheets, statistical software, and many calculators can quickly calculate r. The correlation coefficient r is the bottom item in the output screens for the LinRegTTest on the TI-83, TI-83+, or TI-84+ calculator (see previous section for instructions).

The Coefficient of Determination

The variable r^2 is called the coefficient of determination and is the square of the correlation coefficient, but is usually stated as a percent, rather than in decimal form. It has an interpretation in the context of the data:

- r^2, when expressed as a percent, represents the percent of variation in the dependent (predicted) variable y that can be explained by variation in the independent (explanatory) variable x using the regression (best-fit) line.
- $1 - r^2$, when expressed as a percentage, represents the percent of variation in y that is NOT explained by variation in x using the regression line. This can be seen as the scattering of the observed data points about the regression line.

Consider the third exam/final exam example introduced in the previous section

- The line of best fit is: $\hat{y} = -173.51 + 4.83x$
- The correlation coefficient is $r = 0.6631$